ENTERPRISE SYSTEM AUDIT

Database Architecture Report

Full schema audit: 27 tables, relationships, indexes, and data-integrity findings.

Company Safari Zangu Ventures
Engagement Full-Codebase Enterprise Audit
Date 20 September 2026
Document 3 of 6
Classification Internal — Management & Engineering

Methodology: every finding in this report is drawn directly from the live codebase, database schema, and configuration — file paths and line numbers are cited throughout. No functionality is assumed to exist without direct verification.

1. Database Scorecard

DATABASE SCORE **8.5 / 10**

Strong referential integrity (23 real FKs) and indexing; docked for 3 config/DB duplicate-source-of-truth risks and 7 scaffolded-but-unused tables

1.1 At a Glance

Metric	Value	Note
Tables	27 + 1 view	
Enforced foreign keys	23	Verified via information_schema.KEY_COLUMN_USAGE
Tracked migrations	27	Real schema_migrations table, populated
Largest table	analytics_events (967 rows)	Actively used first-party tracking
Orphaned/unused tables	7	See Section 3 below

2. Full Table Inventory

Table	Rows	Purpose	Group
admin_users	2	Staff/agent accounts, role, commission_rate	Core
customers	2	Deduplicated contact record shared across leads/quotations/bookings	Core
leads	1	Sales pipeline (8-stage enum), agent assignment, commission fields	Core
quotations	14	Full itemized quote — 45 columns, richest table in the schema	Core
bookings	13	Confirmed bookings, payment_status, trip_extension	Core
commissions	0	Agent commission ledger, status workflow (pending/approved/paid/cancelled)	Core
packages	6	6 real, fully-populated safari packages	Catalog
itineraries	13	Day-by-day JSON itinerary content, linked from packages	Catalog
accommodations	60	60 real lodges/camps with per-tier rates, single supplement, child rate	Catalog
vehicles	4	4 vehicle types with full cost breakdown (fuel/driver/maintenance/insurance)	Catalog
activities	10	10 paid add-on activities (balloon safari, walking safari, etc.)	Catalog
park_fees	8	8 Tanzania parks, adult/child fee + dormant resident_adult/resident_child columns	Catalog
seasonal_rates	3	Peak/Shoulder/Low global multipliers, actively used	Catalog
accommodation_seasonal_rates	0	Per-accommodation seasonal override table — schema only, never populated	Catalog (unused)
package_variants	0	Tier-variant pricing per package — schema only, never populated	Catalog (unused)
agent_targets	0	Monthly sales targets per agent — schema only, never populated, no admin UI found	Scaffolded, unused
team_members	6	Public-facing About page team bios	Content
contact_messages	7	Public contact form submissions + first_response_at tracking	Inbound
custom_safari_requests	6	Homepage 'Design Your Own' submissions + first_response_at tracking	Inbound
email_queue	12	Outbound email queue/log, attempts/status/last_error	Infra
email_templates	0	Admin-overridable template store — exists, never actually used (0 rows; all 13 templates currently render from the PHP-coded defaults)	Infra (unused)
booking_reminders	15	Scheduled deposit/balance/departure/post-trip reminder emails	Infra

Table	Rows	Purpose	Group
activity_log	0	Admin audit trail (who changed what) — schema built, never actually written to	Infra (unused)
analytics_events	967	First-party page/event tracking — actively used, largest table by row count	Infra
settings	1	Generic key/value settings store, essentially unused (1 row)	Infra (unused)
schema_migrations	27	Migration tracking — real, populated, but several recent .sql files this session weren't registered here	Infra
vw_package_profitability	(view)	SQL VIEW over packages — added this session, operational_cost still 0 for all rows pending real cost data	Derived

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3. Entity Relationship Overview

23 real, database-enforced foreign key constraints connect the schema (not just indexed columns — every relationship below is a true FOREIGN KEY with referential integrity, verified against information_schema, not inferred from naming conventions).

Table	Foreign Keys
bookings	customer_id → customers.id, package_id → packages.id, itinerary_id → itineraries.id, quotation_id → quotations.id
quotations	customer_id → customers.id, package_id → packages.id, vehicle_id → vehicles.id, itinerary_id → itineraries.id, lead_id → leads.id, booking_id → bookings.id, custom_request_id → custom_safari_requests.id, created_by → admin_users.id
leads	customer_id → customers.id, agent_id → admin_users.id, assigned_to → admin_users.id
commissions	agent_id → admin_users.id, lead_id → leads.id, approved_by → admin_users.id
custom_safari_requests	customer_id → customers.id
packages	itinerary_id → itineraries.id
agent_targets	agent_id → admin_users.id
activity_log	actor_id → admin_users.id

Not formally FK-constrained (a deliberate, reasonable choice in most cases): commissions.booking_ref and bookings.booking_ref are linked by a business key (varchar) rather than an integer FK — standard practice when the reference needs to remain valid even if a row is later archived, but worth confirming the application layer never lets these drift out of sync.

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4. Duplicate Data Sources

Three cases were found where the same business data is defined in both a config/app.php constant AND a live database table. In all three, the application code correctly treats the database as the source of truth and only falls back to the config constant inside a catch block if the database query fails — this is a defensible defensive-coding pattern, not an active bug. The risk is specifically about what happens in that failure case.

PARK_FEES config array vs park_fees DB table — values have drifted

MEDIUM

config/app.php defines a PARK_FEES constant (Serengeti adult=$70, Ngorongoro adult=$70) that duplicates the park_fees DB table (Serengeti adult_fee=$82, Ngorongoro adult_fee=$80). App\Services\QuotationEngine::getParkFee() correctly reads the DB table first and only falls back to the stale config constant inside a catch block (QuotationEngine.php:201-217) — so this is NOT an active pricing bug today. It IS a real risk: if the DB connection ever fails mid-quote, the fallback would silently undercharge by $10-12 per park entry using 2+-generation-old rates, with no alert that it happened.

```
EVIDENCE: config/app.php:151-159 vs `SELECT * FROM park_fees` (live);
QuotationEngine.php:201-217
```

SEASONAL_RATES config array vs seasonal_rates DB table — currently in sync

LOW

Same DB-first/config-fallback pattern as park fees, but the values currently match exactly (peak=1.0/shoulder=0.85/low=0.75 in both). Same latent-drift risk as above, just not triggered yet.

```
EVIDENCE: config/app.php:180-182; seasonal_rates table; QuotationEngine.php:168-174
```

ACTIVITY_PRICES config array vs activities DB table — currently in sync

LOW

Same pattern again, all 10 activity prices verified matching between config and DB today.

```
EVIDENCE: config/app.php:185-197; activities table; QuotationEngine.php:247-256
```

package_costs static config vs packages.cost_breakdown JSON INFO

Not a live conflict — package_costs_seed.sql / package_costs_upgrade.sql were one-time seeding scripts for the packages.cost_breakdown JSON column, not a second live pricing source.

`EVIDENCE: sql/package_costs_seed.sql, sql/package_costs_upgrade.sql`

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5. Schema Built, Never Wired Up

Seven tables/columns exist with real schema (and in several cases a full admin CRUD interface) but are never read by any other part of the application — confirmed via exhaustive cross-reference search, not inference from naming.

Table / Column	Status
park_fees.resident_adult / resident_child	Columns exist, all 8 rows are $0.00, and QuotationEngine.php's SELECT never queries them — resident/EAC pricing is schema-ready but not implemented anywhere in calculation, UI, or admin editing.
accommodation_seasonal_rates	Table created, 0 rows, no admin page manages it, nothing reads from it — a per-accommodation seasonal override layer that was scaffolded but never wired up.
package_variants	Table created, 0 rows, no admin page manages it — tier-variant pricing per package was scaffolded but never wired up.
agent_targets	Table created, 0 rows, no admin page reads or writes it — sales-target tracking was scaffolded but has no UI at all.
email_templates (is_custom flag)	The admin-editable-template override system exists in code (EmailTemplate::render() checks this table first) but the table has 0 rows — every email currently renders from hardcoded PHP, not the DB override path. admin/email_templates.php exists as a page but nothing has been saved through it.
activity_log	A full audit-log table + App\Services\ActivityLog service class exist, but the table has 0 rows — nothing in the admin panel is currently calling ActivityLog::record() despite the infrastructure being built.
settings	A generic key/value settings table exists with exactly 1 row — essentially dead infrastructure.

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